

Matching the Statistic to the Hypothesis: Geometric-Null Binomial Proximity Testing for Harmonic Longitude Structure in Monument Distributions

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Abstract

Claims that ancient or sacred monuments align with longitude grids raise two separable questions: whether a spatial pattern is statistically detectable, and how such a pattern should be interpreted archaeologically. Following [Silva \(2020\)](#), this paper treats the first as a problem of statistical description and the second as a separate historical, cultural, and geographic task. The methodological focus is statistical matching. The Rayleigh periodogram, a standard circular-statistical diagnostic in archaeoastronomy, measures mean phase concentration at period T . A thresholded repeated-line hypothesis, however, predicts proximity to any harmonic longitude within a specified tolerance. These are distinct quantities. Rayleigh R remains useful for detecting circular phase concentration, but it can dissociate from harmonic proximity: a concentrated off-harmonic cluster can produce large R , while many near-harmonic sites mixed with off-harmonic mass can yield proximity enrichment with weak Rayleigh concentration.

The matched statistic for “sites cluster near harmonics of period T ” is a one-sided binomial count of sites within distance d of any harmonic, benchmarked against the geometric null $p_0 = 2d/T$. The framework is demonstrated under a fixed operational design: fixed period, fixed corridor phase, fixed thresholds, and fixed monument corpora, including a UNESCO morphology-term keyword corpus defined by words denoting dome-like or hemispherical architecture. In the UNESCO dome/stupa corpus ($N = 90$), a period sweep shows the Rayleigh–proximity dissociation directly: Rayleigh significance appears at 2.4° , 3.5° , and 3.6° without corresponding proximity enrichment, while the enriched 3.0° period is Rayleigh-non-significant. Applied to three operational hemispherical-monument corpora at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase, the geometric-null binomial tests identify a consistent corpus-level harmonic-proximity pattern, summarized descriptively by Fisher’s method ($p < 0.001$ at Tier A). The contribution is a matched statistical framework for thresholded harmonic-proximity hypotheses and an empirical demonstration that Rayleigh concentration and harmonic proximity are complementary but non-interchangeable descriptors of spatial structure.

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1 Introduction

Do monuments cluster near specific longitude intervals? The question recurs in the archaeoastronomical and archaeogeographic literature, usually motivated by observational coincidences involving sacred sites, trade corridors, or cosmographic traditions. It has also been criticized on grounds of investigator-assembled site lists, insufficient correction for multiple comparisons, and lack of independent replication (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). These concerns motivate a stricter separation between statistical detection and archaeological explanation. Alongside them, this paper raises a complementary issue of statistical matching: the standard circular-statistical tool used in this literature measures a different quantity than the one a repeated-line grid hypothesis predicts.

The tool in question is the Rayleigh statistic R . R measures the degree to which a set of phases concentrates toward a single preferred direction on the unit circle — it is a sensitive test for unimodal directional concentration. Prior studies have applied Rayleigh tests to longitude distributions and interpreted a significant R at some period T as evidence that sites are placed near the harmonics of T (Thom, 1967; Magli, 2013, 2016).

This is a useful diagnostic but only a partial answer to the repeated-line question. A grid-placement hypothesis predicts that sites fall near longitudes $\lambda_0, \lambda_0 + T, \lambda_0 + 2T, \dots$. Under modulo- T phase mapping these harmonics collapse to a common phase, so an ideal grid-aligned corpus would produce strong Rayleigh concentration. In real geographically clustered corpora, however, Rayleigh concentration and thresholded harmonic proximity can dissociate: a dominant off-harmonic cluster can produce significant Rayleigh concentration, while a corpus with many near-harmonic sites plus substantial off-harmonic mass can produce binomial enrichment without significant Rayleigh R . A non-significant Rayleigh result therefore does not rule out harmonic proximity, and a significant Rayleigh result does not by itself establish proximity to repeated longitude lines. The two statistics are complementary; they answer related but distinct questions.

The directly matched test for “sites cluster near harmonics” is straightforward. Under a uniform longitude model, the expected fraction of longitudes falling within $\pm d^\circ$ of any harmonic of period T is $p_0 = 2d/T$. This is the *geometric null*. In clustered corpora it remains a useful geometric benchmark, but the resulting p -values are interpreted here as corpus-level proximity enrichment rather than counts of independent placement decisions.

This paper develops that matched test and uses the monument data as a worked methodological example. Under a fixed period, fixed corridor phase, fixed proximity thresholds, and fixed operational corpora, the geometric-null binomial test estimates whether more sites fall near harmonic longitudes than expected from the geometric coverage of the threshold bands. Rayleigh statistics are retained as complementary circular diagnostics, and the period sweep is used to show when phase concentration and thresholded harmonic proximity agree or dissociate. The empirical application identifies a reproducible corpus-level proximity-enrichment pattern at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Following Silva (2020), this is treated as statistical detection

and description of spatial structure; historical, cultural, or geographic explanation is a separate interpretive task requiring independent evidence.

2 Rayleigh Concentration and Harmonic Proximity: Complementary but Distinct Statistics

2.1 Why the Two Statistics Can Dissociate

Let $\lambda_1, \dots, \lambda_N$ be site longitudes and T a candidate grid period with anchor λ_0 . Define phases by $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$, and the Rayleigh statistic by

$$R = \left| \frac{1}{N} \sum_{i=1}^N e^{i\theta_i} \right|.$$

A key property of this transformation is that sites near *any* harmonic of T — near λ_0 , near $\lambda_0 + T$, near $\lambda_0 + 2T$, and so on — all map to the same phase $\theta \approx 0$ under the modulo- T reduction. Under ideal conditions (all sites falling exactly on a harmonic), R would equal 1. The modulo- T phase map therefore does not, by itself, produce multiple phase classes: harmonics collapse to one class, not many.

The Rayleigh test and the binomial proximity test can nevertheless dissociate because they aggregate cross-site information differently.

What Rayleigh measures. R is the length of the mean phasor: it is sensitive to whether a *single direction* on the unit circle predominates after averaging over the full corpus. In a geographically clustered corpus, a single dominant geographic cluster can produce large R at period T simply because that cluster’s longitude maps to a concentrated phase at that particular T — regardless of whether the cluster is harmonically proximate. This is a generic property of clustered data on the unit circle; it can produce significant R at periods that carry no harmonic-proximity interpretation.

What the binomial measures. The binomial proximity test counts the number of sites whose deviation from the nearest harmonic falls within a threshold d : $\#\{i : \delta_i \leq d\}$. Let $r_i = ((\lambda_i - \lambda_0) \bmod T)$ and $\delta_i = \min(r_i, T - r_i)$. This is independent of the mean phase direction. Sites far from harmonics (large δ_i) contribute zero to the binomial count regardless of whether they push R up or down. Conversely, a corpus can show significant binomial enrichment even when many off-harmonic sites keep the mean phasor small.

Complementarity. The two tests ask complementary questions. Rayleigh asks: does one phase direction dominate? The binomial asks: are more sites within a fixed reporting threshold of a harmonic than the uniform null predicts? In a real corpus these can agree or disagree, depending on the balance between harmonically proximate sites, off-harmonic sites, and the geographic clustering structure of the corpus. A non-significant Rayleigh result does not rule out harmonic proximity; a significant Rayleigh result does not establish harmonic proximity unless the phase concentration is verified to coincide with a harmonic. Relying on Rayleigh alone to evaluate a repeated-line hypothesis can therefore produce false positives at cluster-dominated periods and miss genuine proximity enrichment when near-harmonic sites coexist with substantial off-harmonic mass.

2.2 Empirical Demonstration: Period Sweep Dissociation

Table 1 illustrates the dissociation concretely using the UNESCO keyword-matched dome/stupa corpus ($N = 90$). Four candidate periods are shown, selected to represent the full range of outcomes: two Rayleigh-significant periods without proximity enrichment, one period with proximity enrichment but no Rayleigh significance, and one period with neither. Both tests use the Tier-A threshold ($\pm 0.10 \times T$, geometric null $p_0 = 0.20$); finer-threshold tiers (A+, A++) are reported in the primary results.

Table 1: Rayleigh concentration vs. Tier-A binomial proximity enrichment at four candidate periods, UNESCO dome corpus ($N = 90$, corridor phase $\sim 35.25^\circ\text{E}$). Tier-A threshold = $0.10 \times T$; geometric null $p_0 = 0.20$. The two statistics dissociate across periods: Rayleigh-significant periods need not show proximity enrichment, and the period with the strongest proximity enrichment (3.0°) is Rayleigh-non-significant. This illustrates that the two tests measure different aspects of the phase distribution; neither subsumes the other.

Period	Rayleigh p	Sig	Tier-A hits	Expected	Binom. p	Sig
2.4°	0.0083	**	21	18.0	0.2503	ns
3.0°	0.1796	ns	29	18.0	0.0043	**
3.5°	0.0267	*	13	18.0	0.9312	ns
3.6°	0.0407	*	20	18.0	0.3383	ns

The dissociation is clear. Rayleigh is significant at 2.4° , 3.5° , and 3.6° , and non-significant at 3.0° . The binomial proximity test is significant only at 3.0° . A researcher relying on Rayleigh alone to evaluate the repeated-line hypothesis would flag periods that carry no proximity signal, and would overlook the period with the clearest proximity structure. Using the two tests together avoids this.

The pattern is consistent with the mechanism described above. At 2.4° , 3.5° , and 3.6° , one geographic cluster dominates the phase distribution, producing concentrated R without harmonic proximity enrichment. At 3.0° , many sites fall near the harmonic class, but substantial off-harmonic mass keeps the mean phasor small; the binomial test correctly registers the thresholded per-site proximity counts. Rayleigh and the binomial measure genuinely different aspects of the data; neither is wrong, but only the binomial directly answers the question “do sites fall near harmonics?”

A broader candidate-period survey of the dome/stupa corpus (1.5° to 6.0° in 0.1° steps, 46 periods; four representative periods shown in Table 1) finds the dominant binomial proximity peak at 3° ($p = 0.004$); selected nearby non-harmonic spacings (1.8° , 2.1° , 2.7° , 3.3°) are non-significant in the same sweep. This motivates 3° as the primary period for the cross-corpus analysis in Section 5.

A Monte Carlo spacing test (100,000 permutations, dome-label shuffle) evaluates whether the number of significant proximity peaks across 18 pre-specified candidate spacings spanning 0.5° to 15° is surprising under geographic concentration alone. The observed count of 4 significant spacings is marginal ($p \approx 0.098$). Of the four significant spacings, three (6° , 8° , 12°) are multiples of 3° : a node-overlap analysis confirms that all of their hits are accounted for by the 3° harmonic structure, so their off-node components are non-significant. The 3° spacing is the only one whose significance is not explained by overlap with another spacing’s nodes.

3 The Geometric-Null Binomial Framework

3.1 The Geometric Null

For a grid of period T anchored at λ_0 , define the anchor-relative remainder and nearest-harmonic deviation as

$$r_i = ((\lambda_i - \lambda_0) \bmod T), \quad \delta_i = \min(r_i, T - r_i).$$

A site falls within tier threshold d if $\delta_i \leq d$. Under a uniform longitude model, the expected fraction of longitudes satisfying that inequality is $p_0 = 2d/T$. In clustered corpora this remains a useful geometric benchmark, but the resulting p -values are interpreted here as corpus-level proximity enrichment rather than counts of independent placement decisions.

Geographic clusters, serial nominations, and shared cultural traditions are present in all three corpora. Retaining them preserves the operational corpus definition and avoids imposing an external, theory-laden decision about the relative importance of major monument nodes versus smaller dependent sites. If a major stupa center generated many associated monuments, that multiplicity is part of the observed historical-geographic pattern being measured. The binomial test therefore evaluates proximity enrichment in the corpus as documented, as a corpus-level measure rather than a count of independent intentional placement decisions.

The two null models reported in Section 5 (Null A: within-dome bootstrap, reported for localization and transparency; Null B: same-footprint UNESCO background, the discriminating comparison) provide complementary assessments of the UNESCO signal; they contextualize the p -values without requiring a theory of why sites are geographically distributed as they are.

3.2 Tier Classification

We define three proximity thresholds as round fractions of the $T/2$ half-period (1.5°), calibrated to bracket the empirical natural proximity scales of the corpora under study: the OWTRAD Silk Road network (2.6%), the UNESCO dome subset (3.6%), the full UNESCO Cultural/Mixed corpus (10.2%), and the Wikidata stupa corpus (20.2%). Each threshold corresponds to a geometric null rate $p_0 = 2d/T$:

Tier	Threshold	Geometric null p_0
A++	$\delta \leq 0.06^\circ$ ($\lesssim 6.66$ km equatorial)	4%
A+	$\delta \leq 0.15^\circ$ ($\lesssim 16.6$ km)	10%
A	$\delta \leq 0.3^\circ$ ($\lesssim 33$ km)	20%

Tier labels are descriptive shorthand; the primary test is always the binomial.

3.3 Cross-Corpus Consistency via Fisher Combination

For each corpus c at tier d , a one-sided binomial test yields p_c . Fisher’s combination statistic $\chi^2 = -2 \sum_c \ln p_c$, referred to $\chi^2(2C)$ for C corpora, summarizes whether the directional excess is consistent across all corpora simultaneously. Fisher’s method formally assumes independent p -values. The three corpora were assembled by distinct processes

and their geographic clusters are substantially non-overlapping, but they share historical and geographic structure (Eurasian corridor, stupa tradition, public data sources). The combined p-value is therefore used descriptively as a cross-corpus consistency summary. Rayleigh statistics are retained as complementary diagnostics for comparability with prior literature.

4 Data

UNESCO dome/stupa subset ($N = 90$). The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1011 Cultural and Mixed inscribed sites. Seven morphological keywords (stupa/stupas, tholos, dome/domed/domes, spherical) were matched against site names, XML descriptions, and OUV text, yielding $N = 90$ sites. All statistical tests use this keyword-matched population.

Wikidata Q180987 stupa corpus ($N = 229$). All *instance-of: stupa* (Q180987) items in the Wikidata knowledge base, extracted via SPARQL query on the *instance-of: stupa* property. The corpus is geographically dominated by the Java/Sumatra node ($\sim 105\text{--}115^\circ\text{E}$), reflecting the documented historical density of stupa construction around Borobudur (Miksic, 1990).

OSM stupa layer ($N = 259$). OpenStreetMap features with tags `building=stupa`, `historic=stupa`, or `ruins=stupa`, assembled by crowd-sourced tagging. Coverage correlates with mapping effort. Uneven contributor density is a coverage limitation; the OSM corpus is therefore treated as corroborating evidence.

All three corpora were tested against the same fixed candidate frame: period 3° and the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Because open datasets are publicly available, the frame is treated uniformly as fixed for the present analysis across all three corpora. The empirical application therefore functions as a frame-specified descriptive spatial test.

Candidate frame. The candidate reference phase is the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. For reference, this longitude is labeled by Mount Gerizim at 35.269°E , a non-corpus topographic point in the central Levant. Gerizim is not included in the monument corpora. It functions only as a convenient geographic label for the corridor-level phase near Jerusalem and the central Levant. The statistical result is band-invariant: longitudes within the same $x.25^\circ\text{E}$ band produce the same proximity enrichment, and the data cannot distinguish Gerizim from Jerusalem (35.232°E) or from other co-grid longitudes within the same proximity tier.

5 Results

5.1 Primary Result: Cross-Corpus Geometric-Null Test

Table 2 reports the primary statistical results: one-sided binomial tests and Fisher combinations across all three corpora.

All three corpora show consistent directional excess at every tier despite substantially different geographic footprints and assembly processes. Fisher combinations are reported descriptively as cross-corpus consistency summaries, with small nominal combined p-values at all three tiers (see Section 3). The UNESCO A++ result is the strongest individual signal ($3.06\times$), reflecting the distributed character of that corpus; Wikidata and OSM A++ results are individually non-significant, reflecting cluster-level rather than

Table 2: Cross-corpus geometric-null harmonic-proximity enrichment at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. For each corpus and tier, the one-sided binomial test compares the observed number of sites within distance d of any harmonic with the expected count under the geometric benchmark $p_0 = 2d/T$. The resulting p -values are read as corpus-level proximity-enrichment measures under this benchmark, not as counts of independent placement decisions. Fisher’s statistic $\chi^2 = -2 \sum \ln p_i$, referred to $\chi^2(6)$ for the three corpora, is reported descriptively as a cross-corpus consistency summary.

Tier	Corpus	N	Hits	Exp	Fold	p_{binom}	Sig
A++ ($p_0=4\%$)	UNESCO domes	90	11	3.6	$3.06\times$	0.0009	***
	Wikidata stupas	229	11	9.2	$1.2\times$	0.311	ns
	OSM stupas	259	12	10.4	$1.16\times$	0.3433	ns
	<i>Fisher $\chi^2(6) = 18.402$, nominal combined $p = 0.0053$</i>						
A+ ($p_0=10\%$)	UNESCO domes	90	18	9.0	$2.0\times$	0.0032	**
	Wikidata stupas	229	33	22.9	$1.44\times$	0.0212	*
	OSM stupas	259	33	25.9	$1.27\times$	0.089	
	<i>Fisher $\chi^2(6) = 24.017$, nominal combined $p = 0.00052$</i>						
A ($p_0=20\%$)	UNESCO domes	90	29	18.0	$1.61\times$	0.0043	**
	Wikidata stupas	229	70	45.8	$1.53\times$	0.0001	***
	OSM stupas	259	69	51.8	$1.33\times$	0.0059	**
	<i>Fisher $\chi^2(6) = 39.661$, nominal combined $p < 0.001$</i>						

individual-site precision. The Tier A Fisher combination ($p < 0.001$) is the most widely supported result across corpora. The p -values are interpreted as frame-conditional quantities for the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase; the anchor-sweep correction and conditionality discussion are in Section 6.

5.2 Supplementary Rayleigh Phase Concentration

Table 3 reports Rayleigh statistics at $T = 3^\circ$ for comparability with prior literature. The full UNESCO corpus shows no Rayleigh periodicity ($R = 0.0224$, perm- $p = 0.6008$, ns), indicating that the signal is not a generic heritage-list property. Watson’s U^2 and Kuiper tests confirm the dome/stupa phase distribution differs detectably from the full UNESCO corpus ($p = 0.0036$, $p = 0.0013$), supporting subset-specific enrichment relative to the full UNESCO background.

5.3 Negative Controls and Scope

The OWTRAD Silk Road route network ($N = 1674$ nodes) does not independently recover the monument phase in the Normalized Phasor Consensus test ($K = 0.7429$, perm- $p = 0.46532$, ns). Using the 1946 edge midpoints as the proximity sample — one value per route segment, non-deduplicated, the most conservative representation of the network — the binomial test is non-significant at both tiers (A++: $p = 0.0908$, ns; A+: $p = 0.0679$, ns). The route network therefore shows no enrichment under either the directional or the per-segment proximity test when each segment is counted once.

A distinct question is whether *high-centrality* route hubs — cities where many inde-

Table 3: Supplementary Rayleigh phase concentration at $T = 3^\circ$, retained for comparability with prior literature. The primary evidence is Table 2. All results are conditional on the fixed corridor-phase frame.

Corpus	N	R	perm- p	Sig
UNESCO full corpus	1011	0.0224	0.6008	ns
UNESCO dome/stupa	90	0.1365	0.1883	ns
Wikidata Q180987	229	0.1669	0.0016	**
OSM stupa layer	259	0.1073	0.0355	*
<i>Fisher (Wikidata + OSM): $\chi^2(4) = 19.551, p = 0.0006$</i>				

pendent routes converge — fall preferentially near harmonics. Degree-weighting each city by the number of edges it connects ($N = 3892$ node-in-edge occurrences) is one proxy for hub centrality: a city connecting six routes contributes six counts, reflecting its structural prominence in the network. Under this weighting the binomial enrichment is significant (A++: $p < 0.0001$, $z = 4.53$; A+: $p = 0.0002$): the most network-central cities fall near the 3° harmonic grid more often than the uniform null predicts.

The UNESCO Americas sub-corpus ($N = 145$) shows A+ rate of 9.7%, below the 10% geometric null, indicating no proximity enrichment at the $\sim 35.25^\circ\text{E}$ Levantine corridor phase in the Western Hemisphere.

A type-matched geographic control uses the North American mound corpus ($N = 236$): NHL- and NRHP-registered earthen burial and platform mounds assembled from the NPS National Register database with coordinates resolved via Wikidata (NPS, 2024; Wikidata, 2024). These are hemispherical structures morphologically comparable to the Eurasian corpora but located entirely in the Western Hemisphere, outside any connection to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. At the 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase the mounds corpus shows no proximity enrichment at any tier: A++: 1/236 (0.4%, null 4%, $p = 1.00$, ns); A+: 11/236 (4.7%, null 10%, $p = 1.00$, ns); A: 45/236 (19.1%, null 20%, $p = 0.66$, ns). All three tiers are at or below the geometric null. This supports the interpretation that the Eurasian signal is geographically specific to the Eurasian corridor rather than a generic consequence of applying the same harmonic-proximity test to hemispherical monument corpora.

Null A (within-dome bootstrap, 100,000 trials) resamples from dome/stupa sites' own longitude distribution and recovers the observed A++ count ($p = 0.5493$, ns). Null A is reported for completeness only. Because it resamples from the observed dome/stupa longitude distribution, its expected hit rate is defined by the same geography that carries the observed signal. It is therefore not a substantive alternative null model; its only role is to confirm that the enrichment is localized to dome-site geography rather than introduced by the full-corpus comparison.

Null B (same-footprint restricted draw, 100,000 trials) asks a narrower subset-specificity question: among all UNESCO cultural sites whose longitude falls within $\pm 5^\circ$ of any dome site longitude, does the dome/stupa subset show greater harmonic proximity than the broader background? The broader same-footprint UNESCO background expects only ~ 5.1124 A++ hits versus the observed 11 ($p = 0.0129$ at the $\pm 5^\circ$ window, surviving Bonferroni correction for $k = 3$ windows tested, $\alpha_{\text{adj}} = 0.0167$), supporting subset-specific enrichment within that geography relative to the full UNESCO background. Nulls A and B are discussed further in Section 6.

5.4 Anchor Concordance

All three corpora recover the same $\sim 35.25^\circ\text{E}$ Levantine corridor phase. The Normalized Phasor Consensus statistic K — which compares normalized mean phasors at the candidate period, independent of tier counts — yields $K \geq 0.9695$ for all pairings of the three monument corpora (Table 4). The three-corpus combination reaches $\text{perm-}p = 0.02437$ with consensus anchor at 0.029° from 35°E . Negative controls show K drops sharply when the monument corpora are replaced with geographic or route-network controls.

Table 4: Selected NPC anchor-concordance configurations at $T = 3^\circ$ (100,000 permutation draws). High K with non-significant two-corpus $\text{perm-}p$ reflects shared Eurasian geography. Negative controls (Rows 4–5) indicate K is not a generic Eurasian property.

Role	Corpora	K	$\text{perm-}p$	Note
Non-nested pair	Dome + Wikidata Stupa	0.9675	0.16287	Primary non-nested
Conservative	UNESCO full + Stupa	0.982	0.12243	Non-nested
Three-corpus	Dome + Stupa + OSM	0.972	0.02437	All 3 monument corpora
Neg. control	UNESCO + Americas + Stupa	0.6723	0.29819	Geographic control
Contrast corpus	UNESCO + OWTRAD	0.7429	0.46532	Route network
Type control	N. American mounds ($N = 236$)	—	—	No proximity enrichment

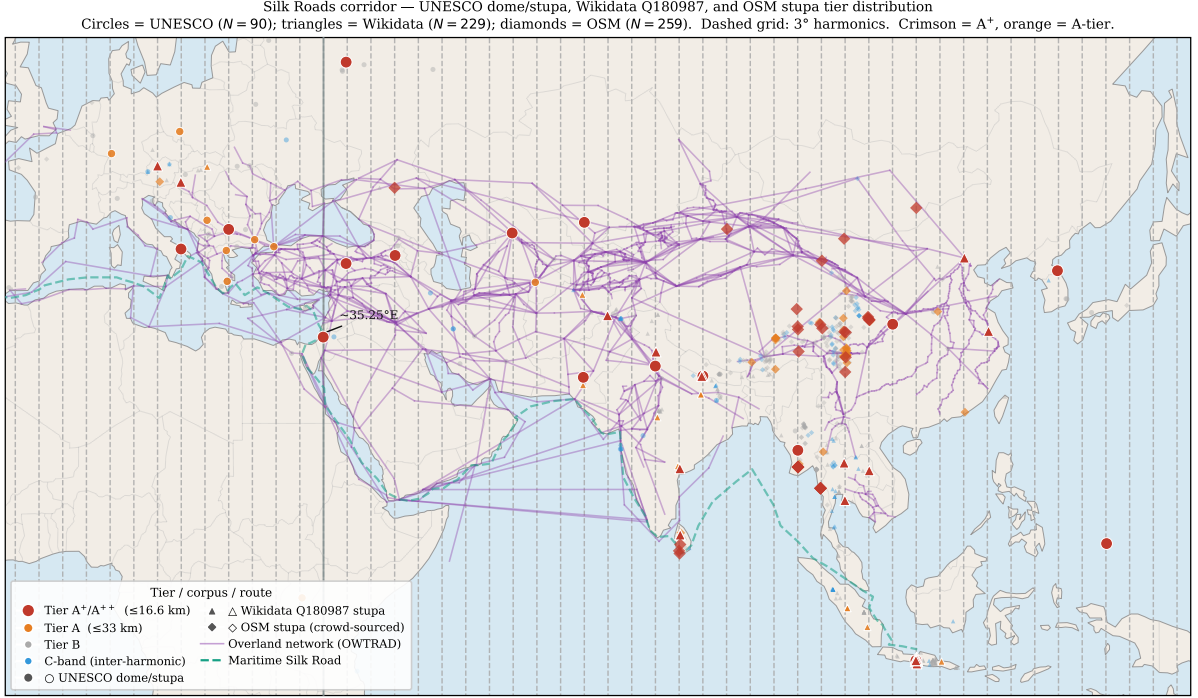


Figure 1: Silk Roads corridor showing all three monument corpora mapped against the 3° harmonic grid (dashed lines) anchored to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Circles = UNESCO dome/stupa ($N = 90$, keyword-matched); triangles = Wikidata Q180987 stupa ($N = 229$); diamonds = OSM stupa ($N = 259$, crowd-sourced). Color indicates harmonic proximity tier: crimson = Tier A⁺ or better (≤ 16.6 km), orange = Tier A (≤ 33 km), gray/blue = B/C-band. The dark meridian marks the fixed $\sim 35.25^\circ\text{E}$ corridor phase. All results are conditional on the fixed corridor-phase frame.

6 Discussion

6.1 The Methods Contribution

The central methodological result is the separation between circular phase concentration and thresholded harmonic proximity. Rayleigh R describes whether the full phase distribution has a dominant mean direction at period T ; the geometric-null binomial describes whether a corpus contains more sites within a stated operational distance of any harmonic than expected from the fraction of longitude covered by those bands. Because the two statistics weight off-harmonic observations differently, they provide complementary information and can dissociate within the same corpus. This makes Rayleigh a useful diagnostic for phase structure and the geometric-null binomial the matched statistic for thresholded proximity-to-harmonics claims. In Silva’s terms, the tests detect and describe structured features in the spatial data; archaeological, historical, or geographic explanations for those features remain interpretive questions to be evaluated with independent evidence.

The analysis is a descriptive spatial test of harmonic proximity. Geographic clustering is part of the documented corpus pattern, so site-level binomial p-values describe corpus-level proximity enrichment rather than independent cultural decisions. The geometric null $p_0 = 2d/T$ gives a transparent expected hit fraction under uniform longitude, and Fisher’s method is used descriptively as a cross-corpus consistency summary. This framework is

applicable to any spatial periodicity hypothesis of this form regardless of geographic or cultural context.

6.2 Interpretation of the Empirical Illustration

The three-corpus demonstration finds consistent directional excess over the geometric null at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase (Fisher $p < 0.001$ at Tier A; this is a frame-conditional proximity-enrichment measure, see Limitations). The UNESCO corpus is the most geographically distributed result, with A-tier sites spanning the full Eurasian corridor.

Null A and Null B address the two tractable aspects of what the UNESCO signal means. Null A (within-dome bootstrap) is reported for completeness only. Because it resamples from the observed dome-site positions, its expected hit rate is defined by the same geography that carries the observed signal — it is not a substantive alternative null model. Its sole informational content is confirming the enrichment is localized to dome-site geography, not an artifact of the comparison procedure ($p = 0.5493$, ns).

Null B provides the discriminating comparison: among all UNESCO cultural sites occupying the same longitude footprint, is the dome/stupa subset more harmonically proximate than the broader background? The answer is yes ($p = 0.0129$, surviving Bonferroni correction), supporting subset-specific enrichment within the geography. The cross-corpus phase concordance (NPC $K \geq 0.9695$ across all pairings of three independently assembled monument datasets with different geographic centers of mass) provides the strongest evidence that the signal is not an artifact of any single corpus’s geographic distribution: UNESCO dome sites, Wikidata Q180987 stupas, and OSM stupa tags converge on the same $\sim 35.25^\circ\text{E}$ phase independently of one another, a convergence that is not explained by any single corpus’s clustering structure alone.

Corpus overlap and curation. The three corpora are assembled by distinct processes: the UNESCO dome/stupa subset derives from keyword matching against the WHC XML dataset; Wikidata Q180987 was extracted via SPARQL query on the *instance-of: stupa* property; OSM was assembled by crowd-sourced tagging of `building=stupa`, `historic=stupa`, and `ruins=stupa` features via Overpass API. A best-effort empirical check using GPS proximity (threshold < 1 km) and name-string matching finds 4 near-coincident site pairs between UNESCO and Wikidata, 0 between UNESCO and OSM, and 1 between Wikidata and OSM (see supplementary overlap audit). Because all three datasets are publicly available, the candidate frame is handled as fixed across the empirical demonstration. The Fisher combination is used as a consistency check across differently assembled datasets. The geographic clusters of the three corpora are substantially non-overlapping at the region level: UNESCO dome sites span the full Eurasian corridor from the Levant to Java; Wikidata is dominated by the Java/Sumatra node; OSM coverage reflects crowd-sourced mapping density across South and Southeast Asia. A single shared geographic cluster does not by itself describe enrichment in all three simultaneously.

The Wikidata result is substantially carried by the Java/Borobudur node, which is historically the dominant node of the stupa tradition (Miksic, 1990): the cluster’s position on a 3° harmonic is the signal the geometric-null test is designed to detect. The OSM result, assembled by a process with no relationship to UNESCO or Wikidata curation, shows the same directional excess ($p = 0.0059$ at Tier A) and the same corridor phase ($K = 0.9695$ with Wikidata). Figure 1 shows all three corpora mapped against the 3°

harmonic grid.

The interpretation follows [Silva \(2020\)](#)’s distinction between statistical structure and archaeological explanation. Significance tests evaluate evidence against a null model; archaeological explanation is developed from evidence independent of the orientations themselves, including the wider archaeological record and socio-historical context. The statistical pattern documented here is compatible with several mechanisms. These include geographic channeling by trade routes and river systems following the Eurasian corridor, cultural diffusion patterns concentrating stupa construction at historically dominant nodes ([Sen, 2014](#)), and selection effects in the UNESCO inscription process. The hub-centrality result in Section 5 is consistent with this: if major route hubs and hemispherical monuments are both concentrated along the same interior Eurasian corridor, the harmonic proximity of each would be a shared consequence of corridor geography rather than an independent signal. Disentangling monument proximity from hub proximity would require a geographic control outside the corridor, which remains a target for future comparative work. Historical interpretations involving deliberate placement would additionally require evidence for pre-modern continental longitude determination at sub-degree precision ([Hunger & Steele, 2019](#); [Neugebauer, 1975](#)). The contribution of the present work is to establish what the directly matched statistical test reports when applied to three differently assembled corpora.

6.3 Limitations

All p -values in Table 2 are conditional on the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Because all three datasets are publicly available open data, it is not possible to determine when or in which corpus the frame was first noticed; we therefore treat all three corpora uniformly as frame-specified rather than attempting to draw a conditional/independent distinction between them. An anchor-sweep correction places the Levantine corridor in the top 2.3% of 36,000 trial anchors globally ($p = 0.0753$, marginal), providing a boundary condition on phase-free discovery claims.

The Bonferroni correction in Null B ($k = 3$, $\alpha_{\text{adj}} = 0.0167$) covers the three window widths ($\pm 2^\circ$, $\pm 5^\circ$, $\pm 10^\circ$) tested in that null. No additional proximity tests were run within Null B beyond these three windows.

7 Conclusion

This paper clarifies the statistic appropriate to a repeated-line longitude hypothesis. Rayleigh concentration and harmonic proximity are related but distinct descriptors of a modulo- T phase distribution: Rayleigh summarizes mean circular concentration, while the geometric-null binomial tests thresholded proximity to any harmonic under the benchmark $p_0 = 2d/T$. The period sweep shows this distinction empirically, with Rayleigh-significant periods lacking proximity enrichment and the enriched 3.0° period lacking significant Rayleigh concentration. Applied under a fixed operational frame — fixed period, corridor phase, thresholds, and corpora — the binomial framework identifies a consistent corpus-level proximity pattern across three hemispherical-monument datasets at the 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Following [Silva \(2020\)](#), the result is best stated as statistically detectable and archaeologically interpretable: the statistical tests describe spatial structure, while questions of mechanism, intentionality, and historical transmission require independent archaeological and socio-historical evidence.

The paper’s contribution is the matched geometric-null framework, the demonstrated Rayleigh–binomial dissociation, and a reproducible empirical pattern that future comparative work can evaluate.

All data, code, and audit files are openly available at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>).

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Data Availability

All data, code, and audit files: Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>). UNESCO XML: UNESCO World Heritage Centre 2024.

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Conflict of Interest

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